

OUTLINE

ENGINE SPECIFICATION [WL-C, WE-C]

dcf010002000104

Specification

Item			Specifications	
			WL-C	WE-C
MECHANICAL				
Type			Diesel, 4-cycle	
Cylinder arrangement and number			In-line, 4 cylinder	
Combustion chamber			Direct injection	
Valve system			DOHC, timing gear and belt driven, 16-valve	
Displacement (ml {cc, cu in})			2,499 {2,499, 152.4}	2,953 {2,953, 180.2}
Bore x stroke (mm {in})			93.0 x 92.0 {3.66 x 3.62}	96.0 x 102.0 {3.78 x 4.02}
Compression ratio			18	18
Compression pressure (kPa {kgf/cm ² , psi} [rpm])			2,900 {29.6, 420.7} [250]	2,900 {29.6, 420.7} [250]
Valve timing	IN	Open BTDC ()	10	10
		Close ABDC ()	30	30
	EX	Open BBDC ()	40	40
		Close ATDC ()	8	8
Valve clearance [engine cold]	IN	(mm {in})	0.10—0.16 mm {0.0040—0.0062 in}	0.10—0.16 mm {0.0040—0.0062 in}
	EX	(mm {in})	0.17—0.23 mm {0.0067—0.0090 in}	0.17—0.23 mm {0.0067—0.0090 in}
LUBRICATION SYSTEM				
Type			Force-fed type	
Oil pressure (reference value) [after warm up] (kPa {kgf/cm ² , psi} [rpm])			410—570 {4.19—5.81, 59.6—82.6} [2,500]	
Oil pump	Type		Gear type	
	Relief valve opening pressure (reference value) (kPa {kgf/cm ² , psi})		580—700 {5.9—7.1, 84.1—101.5}	
Oil cooler	Type		Water-cooled	
Oil filter	Type		Full-flow, paper element	
	Bypass pressure (kPa {kgf/cm ² , psi})		80—120 {0.82—1.22, 11.6—17.4}	
Oil capacity (approx. quantity)	Total (dry engine) (L {US qt, Imp qt})		8.0 {8.5, 7.0}	
	Oil replacement (L {US qt, Imp qt})		6.8 {7.2, 6.0}	
	Oil and oil filter replacement (L {US qt, Imp qt})		7.0 {7.4, 6.2}	
COOLING SYSTEM				
Type			Water-cooled, forced circulation	
Coolant capacity (approx. quantity) (L {US qt, Imp qt})			9.4	
Water pump	Type		Centrifugal, V-belt driven	
Thermostat	Type		Wax, bottom-bypass	
	Opening temperature (C { F})		80—84 {176—183}	
	Full-open temperature (C { F})		95 {203}	
	Full-open lift (mm {in})		8.5 {0.33} or more	
Radiator	Type		Corrugated fin	
Cooling system cap	Cap valve opening pressure (kPa {kgf/cm ² , psi})		93.2—122.6 {0.95—1.25, 13.5—17.8}	
Cooling fan	Type		Thermo-modulation type	
	Number of blades		9	
	Outer diameter (mm {in})		450 {17.7}	
INTAKE AIR SYSTEM				
Turbocharger type			Variable geometry turbocharger	

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Air cleaner element		Dry type	
Glow plug type		Stainless type	
FUEL SYSTEM			
Supply pump		Electronic control	
Fuel injector		Electromagnetic control	
Fuel tank capacity (reference) (L {US gal, Imp gal})		70 {18, 15}	
EMISSION SYSTEM			
EGR valve type		Vacuum	
Catalytic converter type		Oxidation catalytic converter	
CHARGING SYSTEM			
Battery	Voltage (V)	12	
	Type and capacity (5-hour rate) (A·h)	65D31R (56)	95D31R (64)
Generator	Output (V-A)	12-70	
	Regulated voltage (V)	14.1—14.7	
	Self diagnosis function	Equipped	
STARTING SYSTEM			
Starter	Type	Coaxial reduction	
	Output (kW)	2.2	
CONTROL SYSTEM			
IAT sensor No.2 (Inside MAF)		Thermistor	
MAF sensor		Hot-wire	
IAT sensor No.1		Thermistor	
Boost sensor		Piezoelectric element	
ECT sensor		Thermistor	
CMP sensor		Hall element type	
CKP sensor		Magnetic pickup	
APP sensor		Potentiometer	
EGR valve position sensor		Potentiometer	
BARO sensor (built into PCM)		Piezoelectric element	
Fuel temperature sensor		Thermistor	
Fuel pressure sensor		Piezoelectric element	
Neutral switch		ON/OFF	
CPP switch		ON/OFF	
Idle switch		ON/OFF	

Recommended engine oil

Item	Specification
Grade	API CD/CE/CF-4 or ACEA B1/B3/B5
Viscosity (SAE)	5W-30, 10W-30